

L6b: Single Index Models, Portfolio Allocation, and a Risk-Free Asset

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you should be able to:

- Interpret and estimate a single index model (SIM): explain how a factor model describes the co-movement of growth rates, interpret alpha, beta, and the residual, and estimate alpha and beta with their uncertainty and the model's fit.
- Build minimum-variance portfolios from SIM inputs: derive the SIM mean vector and covariance matrix, identify the co-movement the model leaves out, and compare the allocations with data-driven ones. How different are they, and why?
- Add a risk-free asset: derive the capital allocation line and the tangent portfolio, and explain how lending, borrowing, and financing limits affect portfolio choice.

Lectures and Examples

Lecture: Single Index Models, Portfolio Allocation, and a Risk-Free Asset

From L6a: Portfolio Wealth with Multiple Asset GBM
Simulate buy-and-hold wealth and estimate the chance of beating a target NPV.

Interview: Choose the Firms with a Client Interview
The answers set a beta band and a risk-free fraction, and a fixed rule picks the firms.

Example 1: Compare Data-Driven and SIM Portfolios
Compare the covariances, frontiers, and weights, then test on 2025 prices.

Example 2: Construct the Tangent Portfolio and Capital Allocation Line
Does every target pick the same risky portfolio? Test lending and borrowing in 2025.

Review on your own: Estimate Single Index Models from Historical Data and Estimate Parameter Uncertainty by Simulation, which supply the SIM parameters.

Concept Review: Estimated Inputs

Which estimated inputs does each L6a allocation depend on?

- The GMV weights use only the estimated covariance $\hat{\boldsymbol{\Sigma}}_g$. A growth target or floor also uses the mean growth rates $\mathbf{g}'$, which are harder to pin down.
- The optimizer treats its inputs as exact, so it favors asset combinations whose variance is underestimated or whose mean growth is overestimated.

We therefore evaluate allocations on prices that were not used to choose them.

Today, Examples 1 and 2 test the SIM and risk-free allocations on 2025 prices.

Concept Review: Wealth and Scaled NPV

Buy $n_i = w_i W_0 / S_i(0)$ shares, with $w_i \geq 0$ and $\sum_i w_i = 1$, and hold them fixed. For a sale after T years at constant benchmark growth rate g_y:

$$W_t = W_0 \sum_{i=1}^{M} w_i \frac{S_i(t)}{S_i(0)} \qquad \text{portfolio value in dollars}$$

$$\rho_T = \frac{\mathrm{NPV}(g_y, T)}{W_0} = \frac{W_T}{W_0} e^{-g_y T} - 1 \qquad \text{discounted gain per dollar invested.}$$

The sign of ρ_T shows whether the investment beats or trails the benchmark. We allow fractional shares and omit dividends and trading costs.

The L6a data example applies W_t to the held-out 2025 prices. The L6a MAGBM example simulates wealth with the fitted model to estimate $\mathbb{P}(\rho_T > \rho_\star)$.

From L6a: Portfolio Wealth with Multiple Asset GBM

Company Profile: MSCI

MSCI, originally Morgan Stanley Capital International, builds the indexes, risk models, and data that investors use to measure markets and manage portfolio risk. In 2004 it combined with Barra, the risk-modeling firm Barr Rosenberg founded in 1975.

- Indexes: the MSCI ACWI covers about 85% of the investable stock market in 47 countries. Investors use indexes like this as the market when they measure beta.
- Factor risk models: the Barra models describe how more than 90,000 securities move together using market, country, industry, and style factors.
- Predicting beta: Rosenberg and Guy (1976) showed that a firm's business fundamentals help predict its beta.

Today's connection: the SIM is the simplest factor model. What does one market factor leave out?

Explore: Early careers Factor Modeling (MIT OpenCourseWare)

Factor Models

The minimum-variance problem of L6a needs a covariance for every pair of assets: $|\mathcal{P}|(|\mathcal{P}|+1)/2$ numbers from the same few years of data. Yet most stocks move together when market-wide news arrives. A factor model explains that shared movement with a few common factors:

$$g_{i,t} = \underbrace{\alpha_i}_{\text{firm}} + \underbrace{\beta_i\, g_{M,t}}_{\text{market}} + \underbrace{\sum_{j=1}^{m} \gamma_{ij}\, f_{j,t}}_{\text{other factors}} + \underbrace{\varepsilon_{i,t}}_{\text{residual}}$$

where $g_{i,t}$ and $g_{M,t}$ are the growth rates (year^{-1}) of asset i and the market on day t. The $f_{j,t}$ are m other factors, such as interest rates or inflation, and β_i and γ_{ij} are the asset's sensitivities to the factors.

Two Well-Known Factor Models

In our growth-rate notation:

Capital Asset Pricing Model (CAPM): the market is the only factor, in excess of the risk-free growth rate g_f, with no intercept:

$$g_{i,t} - g_f = \beta_i \left(g_{M,t} - g_f \right) + \varepsilon_{i,t}$$

Fama–French three-factor model: size and value factors join the market:

$$g_{i,t} = \alpha_i + \beta_i \, g_{M,t} + s_i \, \mathrm{SMB}_t + h_i \, \mathrm{HML}_t + \varepsilon_{i,t}$$

where s_i and h_i are the sensitivities to SMB (small minus big firms) and HML (high minus low book-to-market firms). The three factors explain over 90% of the return variation in 21 of 25 test portfolios, versus 2 for the market alone.

We keep only the market factor: the single index model (SIM).

Source: Fama and French (1993)

The Single Index Model

With SPY as the market, for asset $i \in \mathcal{P}$ on trading day t (Sharpe, 1963):

$$g_{i,t} = \underbrace{\alpha_i}_{\text{intercept}} + \underbrace{\beta_i\, g_{M,t}}_{\text{market}} + \underbrace{\varepsilon_{i,t}}_{\text{residual}}$$

Here, β_i is the market exposure (dimensionless) and $\varepsilon_{i,t}$ is the idiosyncratic residual. Growth rates, α_i, and $\varepsilon_{i,t}$ are in year^{-1}. For a single day, the model assumes:

$$\underbrace{\mathbb{E}[\varepsilon_i] = 0}_{\text{zero mean}} \qquad \underbrace{\mathrm{Cov}(\varepsilon_i, g_M) = 0}_{\text{uncorrelated with the market}} \qquad \underbrace{\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \;\; (i \neq j)}_{\text{uncorrelated across assets}}$$

The first two determine α_i and β_i. The third, only the market links two assets, is an extra assumption that the estimation example checks.

What Do the Parameters Mean?

Taking the covariance with g_M and the expectation of both sides of the SIM gives:

$$\mathrm{Cov}(g_i, g_M) = \underbrace{\mathrm{Cov}(\alpha_i, g_M)}_{=0} + \beta_i \underbrace{\mathrm{Cov}(g_M, g_M)}_{=\mathrm{Var}(g_M)} + \underbrace{\mathrm{Cov}(\varepsilon_i, g_M)}_{=0}$$

$$\implies \quad \beta_i = \boxed{\frac{\mathrm{Cov}(g_i, g_M)}{\mathrm{Var}(g_M)} = \rho_{iM} \frac{\sigma_{g,i}}{\sigma_{g,M}}}$$

$$\alpha_i = \mathbb{E}[g_i] - \beta_i \, \mathbb{E}[g_M]$$

where $\sigma_{g,i}$ and $\sigma_{g,M}$ are growth-rate standard deviations and ρ_{iM} is their correlation.

Beta is the slope of the line: the change in the asset's predicted growth rate per unit change in the market growth rate.

Systematic and Idiosyncratic Risk

The risk interpretation of beta comes from the variance of both sides of the SIM:

$$\mathrm{Var}(g_i) = \beta_i^2 \, \mathrm{Var}(g_M) + \mathrm{Var}(\varepsilon_i) + 2\beta_i \underbrace{\mathrm{Cov}(g_M, \varepsilon_i)}_{=0} \qquad (\alpha_i \text{ is constant})$$

$$= \boxed{\underbrace{\beta_i^2 \, \sigma_{g,M}^2}_{\text{systematic}} + \underbrace{\sigma_{g,\varepsilon,i}^2}_{\text{idiosyncratic}}}$$

- Systematic: comes from the market. Adding assets that share the same market does not diversify it away.
- Idiosyncratic: belongs to the asset alone, with residual variance $\sigma_{g,\varepsilon,i}^2 = \mathrm{Var}(\varepsilon_i)$.

Beta as a Measure of Relative Risk

- $\beta_i = 1$: the asset carries the same systematic risk as the market.
- $\beta_i > 1$: the asset carries more systematic risk than the market, so its total variance is also larger.
- $0 < \beta_i < 1$: the asset carries less systematic risk, but its total variance can still be larger when the residual variance dominates.
- $\beta_i < 0$: the asset moves against the market.

In our dataset, the betas run from near zero to a little above two, with a median close to one.

How do we estimate α_i and β_i from data?

Estimating the Parameters

We estimate the SIM one asset at a time by least squares, which minimizes the sum of squared residuals over N trading days:

$$\underbrace{\begin{bmatrix} g_{i,1} \\ \vdots \\ g_{i,N} \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 1 & g_{M,1} \\ \vdots & \vdots \\ 1 & g_{M,N} \end{bmatrix}}_{\hat{\mathbf{X}}} \underbrace{\begin{bmatrix} \alpha_i \\ \beta_i \end{bmatrix}}_{\boldsymbol{\theta}_i} + \boldsymbol{\varepsilon}, \qquad \boxed{\hat{\boldsymbol{\theta}}_i = \arg\min_{\boldsymbol{\theta}} \|\mathbf{y} - \hat{\mathbf{X}}\boldsymbol{\theta}\|_2^2 = \left(\hat{\mathbf{X}}^\top \hat{\mathbf{X}}\right)^{-1} \hat{\mathbf{X}}^\top \mathbf{y}}$$

The estimates are the sample versions of the parameter formulas, $\hat{\beta}_i = \widehat{\mathrm{Cov}}(g_i, g_M)/\widehat{\mathrm{Var}}(g_M)$ and $\hat{\alpha}_i = g_i' - \hat{\beta}_i\, g_M'$, with sample means g_i' and g_M'. The residuals $\mathbf{r} = \mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\theta}}_i$ estimate $\boldsymbol{\varepsilon}$.

Review on your own: Estimate Single Index Models from Historical Data

Parameter Uncertainty and Goodness of Fit

How well does the model fit? The coefficient of determination is the fraction of the asset's growth-rate variation that the market term explains:

$$R^2 = 1 - \frac{\|\mathbf{r}\|_2^2}{\sum_{t=1}^{N} \left(g_{i,t} - g_i'\right)^2}$$

With an intercept, R^2 is the squared sample correlation, an estimate of ρ_{iM}^2.

How precise are the estimates? Assuming the errors are uncorrelated across days with constant variance:

$$s_{g,\varepsilon,i}^2 = \frac{\|\mathbf{r}\|_2^2}{N-2}, \qquad \widehat{\mathrm{Cov}}(\hat{\boldsymbol{\theta}}_i) = s_{g,\varepsilon,i}^2 \left(\hat{\mathbf{X}}^\top \hat{\mathbf{X}}\right)^{-1}$$

The standard errors are the square roots of the diagonal. Each has its parameter's units, and smaller values mean more precise estimates. With thousands of daily observations, a 95% interval is the estimate plus or minus about 1.96 standard errors.

What If We Had Another Dataset?

The bootstrap answers this by simulation:

Initialization: Given $\hat{\mathbf{X}}$, the estimates $\hat{\boldsymbol{\theta}}_i$, the residuals $\mathbf{r}$, and the number of samples K, fit a normal distribution to the residuals.

For each $k = 1, 2, \ldots, K$ **do**:

1. Draw synthetic residuals $\boldsymbol{\varepsilon}^{(k)}$ from the fitted normal distribution.
2. Create synthetic observations: $\mathbf{y}^{(k)} \leftarrow \hat{\mathbf{X}}\hat{\boldsymbol{\theta}}_i + \boldsymbol{\varepsilon}^{(k)}$.
3. Estimate parameters from them: $\hat{\boldsymbol{\theta}}^{(k)} \leftarrow \left(\hat{\mathbf{X}}^\top \hat{\mathbf{X}}\right)^{-1} \hat{\mathbf{X}}^\top \mathbf{y}^{(k)}$.

Compare the spread of $\hat{\boldsymbol{\theta}}^{(1)}, \ldots, \hat{\boldsymbol{\theta}}^{(K)}$ with the standard errors.

Review on your own: Estimate Parameter Uncertainty by Simulation

The Covariance Implied by a SIM

Stacking the SIM gives $\mathbf{g} = \boldsymbol{\alpha} + \boldsymbol{\beta}\, g_M + \boldsymbol{\varepsilon}$, with mean growth rates $\boldsymbol{\mu}_g = \boldsymbol{\alpha} + \boldsymbol{\beta}\, \mu_{g,M}$, where $\mu_{g,M} = \mathbb{E}[g_M]$. On the diagonal:

$$\mathrm{Var}(g_i) = \beta_i^2\, \mathrm{Var}(g_M) + 2\beta_i \underbrace{\mathrm{Cov}(g_M, \varepsilon_i)}_{=0} + \mathrm{Var}(\varepsilon_i) \qquad \text{expand the SIM}$$

$$= \beta_i^2\, \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2 \qquad \text{second assumption}$$

Off the diagonal ($i \neq j$), the second and third assumptions remove three terms:

$$\mathrm{Cov}(g_i, g_j) = \beta_i \beta_j\, \mathrm{Var}(g_M) + \beta_i \underbrace{\mathrm{Cov}(g_M, \varepsilon_j)}_{=0} + \beta_j \underbrace{\mathrm{Cov}(\varepsilon_i, g_M)}_{=0} + \underbrace{\mathrm{Cov}(\varepsilon_i, \varepsilon_j)}_{=0}$$

$$= \beta_i \beta_j\, \sigma_{g,M}^2$$

The SIM Covariance Matrix

Splitting each entry into its market and residual parts:

$$\boldsymbol{\Sigma}_g = \sigma^2_{g,M} \begin{bmatrix} \beta_1^2 & \beta_1\beta_2 & \cdots \\ \beta_2\beta_1 & \beta_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} + \begin{bmatrix} \sigma^2_{g,\varepsilon,1} & 0 & \cdots \\ 0 & \sigma^2_{g,\varepsilon,2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} = \underbrace{\sigma^2_{g,M}\,\boldsymbol{\beta}\boldsymbol{\beta}^\top}_{\text{market}} + \underbrace{\mathbf{D}_g}_{\text{residual}}$$

Each diagonal entry splits into a market part $\beta_i^2\sigma^2_{g,M}$ and a residual part $\sigma^2_{g,\varepsilon,i}$, which is what $\mathbf{D}_g$ holds.

- Fewer parameters: $2|\mathcal{P}|+1$ numbers instead of $|\mathcal{P}|(|\mathcal{P}|+1)/2$. For the S&P 500, about a thousand instead of more than a hundred thousand.
- What is assumed away: covariance between residuals. Most residual correlations are near zero, but some bank pairs exceed 0.6.

Portfolio Risk under a SIM

For fixed weights $\mathbf{w}$, substituting the SIM entries of $\mathbf{\Sigma}_g$ into the portfolio variance gives:

$$
\begin{aligned}
\mathrm{Var}(g_p) &= \textstyle\sum_i \sum_j w_i w_j \,\mathrm{Cov}(g_i, g_j) && \text{portfolio variance (L6a)} \\
&= \textstyle\sum_i w_i^2(\beta_i^2\sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2) + \sum_i \sum_{j\neq i} w_i w_j \beta_i \beta_j \sigma_{g,M}^2 && \text{SIM entries} \\
&= \sigma_{g,M}^2 \textstyle\sum_i \sum_j (w_i\beta_i)(w_j\beta_j) + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{collect market terms} \\
&= \sigma_{g,M}^2 \big(\textstyle\sum_i w_i\beta_i\big)^2 + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{a perfect square} \\
&= \underbrace{\beta_p^2\,\sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\textstyle\sum_i w_i^2 \sigma_{g,\varepsilon,i}^2}_{\text{residual risk}} && \beta_p = \textstyle\sum_i w_i\beta_i
\end{aligned}
$$

Market risk depends on $\mathbf{w}$ only through β_p. Even weights over many assets with uncorrelated, similar-size residuals drive residual risk toward zero.

Estimated SIM Portfolio Inputs

Substituting the estimates for the population quantities gives the SIM inputs:

$$\hat{\mathbf{g}}_{\text{SIM}} = \hat{\boldsymbol{\alpha}} + \hat{\boldsymbol{\beta}}\, g'_M, \qquad \hat{\boldsymbol{\Sigma}}_{g,\text{SIM}} = s^2_{g,M}\, \hat{\boldsymbol{\beta}}\hat{\boldsymbol{\beta}}^\top + \hat{\mathbf{D}}_g$$

where g'_M and $s^2_{g,M}$ are the market's sample mean and variance, and $\hat{\mathbf{D}}_g$ holds the residual variances, all from the training days. Compared with the L6a inputs:

- The SIM mean vector is the sample-mean vector: a least-squares line with an intercept passes through the sample means $\mathbf{g}'$, so $\hat{\mathbf{g}}_{\text{SIM}} = \mathbf{g}'$.
- The SIM covariance drops the residual covariances: off the diagonal, the sample covariance $\hat{\boldsymbol{\Sigma}}_g$ splits as $\hat{\Sigma}_{g,ij} = \hat{\beta}_i \hat{\beta}_j\, s^2_{g,M} + \widehat{\text{Cov}}(\mathbf{r}_i, \mathbf{r}_j)$, and the SIM keeps the first term. On the diagonal, the two differ only by $(N-1)/(N-2) \approx 1$ on the residual variance.

Every residual variance is positive, so $\hat{\boldsymbol{\Sigma}}_{g,\text{SIM}}$ is positive definite and every feasible allocation problem has a unique solution.

The Long-Only Problem with SIM Inputs

SIM-1 is L6a's long-only problem with SIM inputs and a floor $g_\star$ (year^{-1}):

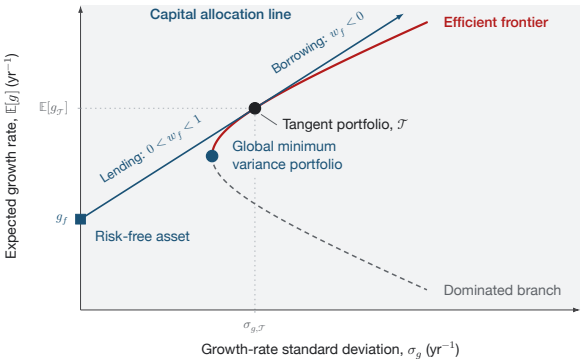
minimize $\mathbf{w}$	$\underbrace{s^2_{g,M}(\hat{\boldsymbol{\beta}}^\top \mathbf{w})^2}_{\text{market risk}} + \underbrace{\sum_i s^2_{g,\varepsilon,i}\, w_i^2}_{\text{residual risk}}$	minimize $\mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{g,\text{SIM}} \mathbf{w}$
subject to	$\hat{\mathbf{g}}^\top_{\text{SIM}} \mathbf{w} \geq g_\star$	meet or exceed the growth target
	$\mathbf{1}^\top \mathbf{w} = 1$	weights sum to one
	$0 \leq w_i \leq 1, \quad i \in \mathcal{P}$	exclude short positions.

This is L6a's LO-1 with a different covariance and the same means. Sweeping $g_\star$ up from the long-only GMV growth rate traces the efficient frontier. We compare both allocations under one covariance, then on 2025 prices.

Examples: Choose the Firms with a Client Interview and Compare Data-Driven and SIM Portfolios

Adding a Risk-Free Asset

A risk-free asset has a known growth rate g_f (year^{-1}) and zero variance over the holding period, such as a Treasury bill held to maturity:



- Capital allocation line (CAL): mixing the risk-free asset with a risky portfolio gives a straight line from $(0, g_f)$ through that portfolio.
- Tangent portfolio $\mathcal{T}$: the steepest such line touches the efficient frontier at $\mathcal{T}$.
- Lending and borrowing: points before $\mathcal{T}$ lend at g_f, and points beyond it borrow at g_f.

The Capital Allocation Line

Put a fraction w_f of wealth in the risk-free asset and $1 - w_f$ in a risky portfolio p with $\sigma_{g,p} > 0$. The complete portfolio c has:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f), \qquad \sigma_{g,c} = |1 - w_f|\,\sigma_{g,p}$$

For $w_f \leq 1$, substituting $1 - w_f = \sigma_{g,c}/\sigma_{g,p}$ gives the capital allocation line:

$$\mathbb{E}[g_c] = g_f + \underbrace{\frac{\mathbb{E}[g_p] - g_f}{\sigma_{g,p}}}_{\text{Sharpe ratio } \mathrm{SR}_p}\ \sigma_{g,c}$$

We lend for $0 < w_f < 1$, hold p alone for $w_f = 0$, and borrow for $w_f < 0$. For daily data, the slope is a one-day Sharpe ratio.

Annualized: Under L5b's model of independent increments, $\mathrm{SR}_p^{\text{annual}} = \sqrt{\tau/\Delta t}\ \mathrm{SR}_p \approx 15.9\ \mathrm{SR}_p$ for $\tau = 1$ year and $\Delta t = 1/252$ years (Sharpe, 1994).

The Tangent Portfolio

The tangent portfolio $\mathcal{T}$ maximizes the Sharpe ratio. Let $\mathbf{\Sigma}_g$ be positive definite, let the GMV portfolio's expected growth exceed g_f, and define $\mathbf{e} = \boldsymbol{\mu}_g - g_f \mathbf{1}$. With short positions allowed, solve:

$$\underset{\mathbf{w}}{\text{maximize}} \quad \frac{\mathbf{e}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \mathbf{\Sigma}_g \mathbf{w}}} \qquad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1 \tag{T-1}$$

At a stationary point, $\mathbf{\Sigma}_g \mathbf{w} = \left(\mathbf{w}^\top \mathbf{\Sigma}_g \mathbf{w} / \mathbf{e}^\top \mathbf{w}\right) \mathbf{e}$, so the weights point in the direction $\mathbf{\Sigma}_g^{-1} \mathbf{e}$. Requiring them to sum to one gives:

$$\boxed{\mathbf{w}_{\mathcal{T}} = \frac{\mathbf{\Sigma}_g^{-1} \left(\boldsymbol{\mu}_g - g_f \mathbf{1}\right)}{\mathbf{1}^\top \mathbf{\Sigma}_g^{-1} \left(\boldsymbol{\mu}_g - g_f \mathbf{1}\right)}} \tag{T-3}$$

With long-only weights, there is no closed form, so we solve numerically.

Two-Fund Separation

The steepest CAL separates choosing a risky portfolio from choosing how much risk to take. Mean-variance investors with the same inputs and risk-free rate hold the same risky fund $\mathcal{T}$, combined with the risk-free asset. Their risk preferences set w_f (Tobin, 1958).

Real investors face limits on how they finance their positions:

- Long-only weights: the risky fund is the long-only maximum-Sharpe portfolio, and investors still choose w_f.
- No borrowing ($w_f \geq 0$): the efficient boundary follows the CAL up to $\mathcal{T}$, then the risky efficient frontier.
- Costlier borrowing: a borrowing rate above g_f gives borrowers a flatter line through a different risky portfolio.

Risky and Risk-Free Allocation with SIM Inputs

Choose the risky weights $\mathbf{w}$ and put the rest, $w_f = 1 - \mathbf{1}^\top \mathbf{w}$, in the risk-free asset. For a target $g_\star > g_f$ (year^{-1}), problem SIM-2 is:

$\underset{\mathbf{w}}{\text{minimize}}$	$\mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{g,\text{SIM}} \mathbf{w}$	minimize growth-rate variance
subject to	$\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w} + \underbrace{\left(1 - \mathbf{1}^\top \mathbf{w}\right)}_{w_f} g_f \geq g_\star$	meet or exceed the growth target
	$0 \leq w_i \leq 1, \quad i \in \mathcal{P}$	exclude short positions.

The fraction w_f has no sign constraint, so $w_f < 0$ means borrowing at g_f.

Eliminating w_f gives the excess-growth constraint $(\hat{\mathbf{g}}_{\text{SIM}} - g_f \mathbf{1})^\top \mathbf{w} \geq g_\star - g_f$.

The Solutions Lie on a Ray

Scaling $\mathbf{w}$ by $c > 0$ scales the excess growth by c and the variance by c^2. So, until a bound $w_i \leq 1$ binds, every target selects the same risky direction:

$$\mathbf{w}(g_\star) = (1 - w_f)\,\mathbf{w}_{\mathcal{T}}, \qquad 1 - w_f = \frac{g_\star - g_f}{\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w}_{\mathcal{T}} - g_f} \tag{SIM-3}$$

where the long-only tangent portfolio, $\mathbf{w}_{\mathcal{T}} = \mathbf{w}/\mathbf{1}^\top\mathbf{w}$, is any solution rescaled to sum to one.

- Targets below $\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w}_{\mathcal{T}}$ lend ($0 < w_f < 1$).
- Targets above it borrow ($w_f < 0$).

The estimation-risk notebook asks how much GMV and tangent allocations move when the estimated inputs change.

Example 2: Construct the Tangent Portfolio and Capital Allocation Line

Optional Advanced Material

These notebooks extend today's material. The advanced index gives a suggested order.

- Residual diagnostics and two bootstrap methods: two bootstraps, heavy tails and autocorrelation, and autocorrelation-consistent standard errors.
- Estimation risk in portfolio weights: resample trading days, compare GMV and tangent weights, and test the long-only ones on 2025 prices.

Summary

- Factor models and the single index model: we kept one market factor to get the SIM. Beta measures systematic risk, and the residual carries idiosyncratic risk. We estimated alpha and beta, their uncertainty, and the model's fit.
- Portfolio inputs from a factor model: we built the covariance from betas and residual variances, with far fewer parameters than a full covariance matrix. The means match, so SIM and data-driven portfolios differ mainly through the dropped residual covariances.
- Risk-free allocation and the tangent portfolio: we derived the capital allocation line and the tangent portfolio, the risky fund with the largest Sharpe ratio. By two-fund separation, investors with the same inputs hold that fund and lend or borrow to set their risk. Financing limits can change the fund.

One market factor supplies the portfolio inputs, and a risk-free asset lets us set the portfolio's risk.